Transfer task: defend and revise

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Learning target: Defend the prediction with evidence and make a visible, justified revision.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 1, Day 19.*
The student multiplied by mass. Divide: $a=F_{net}/m=5$ m/s². Check units N/kg=m/s² or substitute ma back into F_{net}.

02. Worked example: explain the model
Prediction outside interval; possible constant velocity, start position, timing or measurement calibration needs testing.

03. Show the prediction
Central (0,-120)→(16, 11.36). Lower (0,-120.05)→(16, 8.27); upper (0,-119.95)→(16, 14.45). Units million km, millions.

04. Write a public briefing
Use CPA end point (11.36, 35.20) million km; ranges x 8.27-14.45, y 34.4-36 million km. No real impact conclusion from this model.

05. Defend and revise
Evidence-specific feedback and revised calculation/label/reasoning. Accept defended unchanged step with physics justification; no invented peer feedback.

06. Review: connect an earlier idea *Review: Unit 1, Day 10.*
Earth pulls the sign down with 60 N; the cord pulls it up with 60 N.

07. Exit check
Require actual work references. Physically valid claim plus gravity/orbit question. Ratings are self-assessment only.

08. STRETCH: test the model
v=2.5 m/s; predict e.g. 15 m at 6 s and measure. Fitting one observation does not independently validate the model.

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01. Warm-up: bring back an earlier idea *Review: Unit 1, Day 19.*
Central 38 m; half-width 4.5 m; interval [33.5, 42.5] m. Valid checks include units and upper/lower endpoint substitution.

02. Worked example: explain the model
Prediction outside interval; possible constant velocity, start position, timing or measurement calibration needs testing.

03. Graph central and bounded positions
Central start -120 million km, end 11.365912 million km. Bounds initial±0.05 million km; final half-width 3.140963 million km. Width grows with speed uncertainty.

04. Defend the public claim
Use Day 21 Honors answers. No impact probability inferred; gravity, Earth motion and 3 D state excluded. Explain bounds rather than present false certainty.

05. Peer defense and revision
Require specific feedback and justified revision or retained claim. Mathematical and physical reasoning must remain student's own.

06. Review: connect an earlier idea *Review: Unit 1, Day 10.*
Weight 49 N down from Earth and tension 49 N up from string. Object-on-string acts on the string, not the object.

07. Exit check
Specific references in each dimension; revision with support; next question about gravity/orbits. Model does not establish real encounter.

08. STRETCH: test the model
Constant v predicts 15 m, inconsistent. One possible model: 2.5 m/s for first 4 s, then 1.5 m/s for next 2 s. More data needed to distinguish abrupt versus gradual changes.